



An-Najah National University

Faculty of Engineering & Information Technology

Graduation Project :

Automated Egg Incubation System

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Disclaimer statement

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Abstract

Egg incubation is a highly sensitive process that requires precise control of temperature, humidity, ventilation, and egg turning in order to ensure healthy embryo development and high hatching rates. Traditional incubation systems often rely on manual supervision or simple control mechanisms, which may lead to instability, human error, and reduced efficiency.

This project presents the design and implementation of a **Smart Egg Incubator System** based on an embedded control platform. The proposed system utilizes an Arduino Mega microcontroller to monitor environmental conditions and control heating, humidification, ventilation, and egg turning processes automatically. A non-blocking software architecture based on state machines was implemented to ensure reliable real-time operation without interrupting critical system tasks.

Special emphasis was placed on system safety and reliability. Features such as safe boot behavior, controlled egg turning using a stepper motor, emergency stop handling, and protection against unstable power conditions were incorporated. The system was tested under various simulated operating scenarios, including power interruptions and abnormal conditions, and demonstrated stable and predictable behavior.

The results show that the proposed system is capable of maintaining a controlled incubation environment with minimal human intervention, while providing improved safety and robustness compared to basic incubation solutions. This project demonstrates a practical and cost-effective approach to automated egg incubation and serves as a foundation for future enhancements such as real-time clocks, advanced control algorithms, and data logging.

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1.1 Background and Motivation

Egg incubation is a critical and highly sensitive process in which small variations in environmental conditions can directly affect embryo development and hatching success. Maintaining stable **temperature, humidity, air exchange, and regular egg turning** is essential to create the proper incubation environment. In many small-scale settings, incubation is still managed manually or through basic incubators that provide limited control and weak safety behavior during unexpected events such as power interruptions or unstable supply conditions.

The motivation behind this project is to design and implement a low-cost yet reliable automated egg incubator system that reduces human dependency, improves stability, and increases safety. The project focuses on building a practical embedded solution capable of running continuously with predictable behavior under normal operation and under abnormal conditions.

1.2 Problem Statement

Manual incubation or simple control approaches often suffer from several practical limitations. Temperature and humidity may fluctuate due to delayed human intervention or limited control precision. Egg turning may be inconsistent or forgotten, which can negatively impact embryo development. Additionally, sudden power loss can cause unsafe mechanical behavior, especially in systems that rely on motor holding torque to maintain a turning position.

Therefore, the core problem addressed in this project is the need for an incubation system that provides stable environmental control, reliable and smooth egg turning, and safety-oriented behavior that avoids unpredictable actuator motion after power-related events.

1.3 Project Objectives

The main objectives of this project are:

1. To design and implement an automated control system that maintains the incubator environment within suitable temperature and humidity ranges.
 2. To implement a reliable egg turning mechanism using a stepper motor and driver, ensuring smooth motion and predictable positioning.
 3. To develop a non-blocking software architecture based on state machines to guarantee real-time responsiveness without interrupting system tasks.
 4. To incorporate safety-oriented operational behavior such as safe boot, controlled arming of turning actions, and predictable handling of power interruption scenarios.
 5. To validate the system through structured testing scenarios and demonstrate stable and repeatable operation.
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1.4 Project Scope

This project focuses on building a functional prototype of a smart egg incubator using an embedded controller. The system integrates environmental sensing and actuator control for heating, humidification, ventilation, and egg turning. The work includes software design, hardware integration, and validation through practical test scenarios.

The scope of this project does not include live hatching experiments or biological performance evaluation due to time and ethical constraints. Instead, the project evaluates system performance using simulated operating conditions and structured functional tests to verify stability, safety behavior, and correct operation of all subsystems.

1.5 Report Organization

This report is organized as follows: Chapter 2 discusses the project constraints, standards, and relevant coursework background. Chapter 3 reviews related literature on egg incubation systems and automated environmental control. Chapter 4 presents the theoretical concepts that support the system design, including hysteresis control, non-blocking logic, and state machine modeling. Chapter 5 describes the

methodology, including hardware design, software architecture, and system behavior. Chapter 6 presents the tools and technologies used. Chapter 7 provides results and analysis based on system testing. Chapter 8 discusses limitations and lessons learned. Finally, Chapter 9 concludes the report and suggests recommendations and future work.

2.1 Design Constraints

Egg incubation systems intended for small-scale or educational applications are often subject to strict design constraints. In this project, the design was limited by cost considerations, component availability, and development time. The system was required to be built using affordable and commonly available electronic components while maintaining acceptable reliability and functionality. These constraints influenced both the hardware selection and the overall system architecture.

2.2 Environmental and Operational Constraints

The incubator is designed to operate in typical indoor environments where ambient temperature, humidity, and power stability may vary. Power interruptions and voltage fluctuations represent a significant operational constraint, particularly in environments without dedicated backup power systems. The system must therefore handle sudden shutdowns and restarts without causing unsafe or unpredictable behavior.

2.3 Safety and Reliability Constraints

The safety of the incubation process was a key consideration throughout the project. However, due to hardware and time limitations, the system does not implement a fully active emergency recovery mechanism in the event of a power failure. Instead, a safety-oriented design approach was adopted.

Upon power loss, all actuators are deactivated, and the system returns to a safe idle state upon restart. No automatic mechanical motion is resumed without user intervention. While this approach prevents uncontrolled operation after power restoration, it does not actively prevent mechanical back-driving caused by gravity when motor holding torque is lost during power interruption. This limitation was identified as a mechanical constraint rather than a software-related issue.

2.4 Electrical and Mechanical Constraints

From an electrical perspective, the system is constrained by the absence of advanced power conditioning or backup energy storage. Sudden load changes caused by motors and relays may lead to temporary voltage drops, especially during startup or re-arming phases. Mechanically, the egg turning mechanism relies on a stepper motor, which requires continuous power to maintain its position. In the absence of power, the motor may lose holding torque, allowing mechanical components to move freely under gravitational forces.

2.5 Applicable Standards and Design Practices

Although this project was not designed to comply with industrial certification standards, it follows common embedded system design practices related to safety, reliability, and modularity. These practices include separation of high-voltage and low-voltage components, use of non-blocking software structures, controlled actuator sequencing, and clear definition of safe startup and shutdown behavior.

2.6 Relevant Coursework Background

The development of this project was supported by knowledge gained from several undergraduate coursework modules, including Embedded Systems, Microcontrollers, Control Systems, and Digital Electronics. Concepts such as state machine modeling, real-time system behavior, actuator control, and sensor interfacing were directly applied in the system design and implementation.

3.1 Egg Incubation Principles

Egg incubation is a controlled biological process that aims to provide optimal environmental conditions for embryo development. The most critical parameters affecting incubation success are temperature, humidity, ventilation, and egg turning. Deviations from recommended temperature ranges can lead to delayed development, deformities, or embryo mortality. Similarly, improper humidity levels may cause excessive moisture loss or insufficient air cell development. Regular egg turning is necessary to prevent the embryo from adhering to the eggshell and to promote uniform heat distribution.

3.2 Temperature Control in Incubation Systems

Temperature control is the most influential factor in egg incubation systems. Many traditional incubators use simple on/off control mechanisms based on thermostats or basic sensors. While these approaches are cost-effective, they often result in temperature oscillations around the target value. Advanced systems may utilize proportional or PID-based control to improve stability; however, such solutions increase system complexity and cost. In small-scale incubators, hysteresis-based control remains a common and practical solution.

3.3 Humidity Control Methods

Humidity control in incubation systems is commonly achieved using passive water trays, ultrasonic humidifiers, or controlled evaporation mechanisms. Simple systems rely on manual adjustment, while automated incubators monitor humidity using sensors and activate humidification devices as needed. Maintaining stable humidity is challenging due to environmental changes and ventilation effects. As a result, many systems accept moderate humidity fluctuations rather than attempting aggressive control.

3.4 Egg Turning Techniques and Mechanisms

Egg turning mechanisms vary widely in design, ranging from manual tilting trays to motorized rotating frames. Automated systems often employ DC motors, servo motors, or stepper motors depending on required precision and load characteristics. Stepper motors provide precise angular control but rely on continuous power to maintain position. In the event of power loss, systems using stepper motors may experience uncontrolled movement unless mechanical locking mechanisms are implemented.

3.5 Embedded Systems in Incubation Applications

Embedded systems are widely used in modern incubation applications due to their flexibility, low power consumption, and ability to integrate sensing and control functions. Microcontroller-based solutions allow real-time monitoring of environmental parameters and automated control of actuators. However, system reliability depends heavily on software design choices, particularly in handling timing, concurrency, and unexpected operating conditions.

3.6 Reliability and Safety Approaches in Automated Systems

Reliability and safety in automated systems are commonly addressed through a combination of hardware protection and software-based strategies. While some systems implement backup power supplies or mechanical brakes, others adopt fail-safe operational behavior, where the system transitions to a safe state during faults or power loss. The selection of an appropriate approach depends on system complexity, cost constraints, and application risk level.

3.7 Research Gap and Project Contribution

Based on the reviewed literature, many incubation systems focus primarily on environmental control accuracy while providing limited discussion of software architecture and safety-oriented behavior during abnormal operating conditions. This project contributes by emphasizing structured non-blocking software design, state machine-based control, and clear definition of safe behavior during power-related events, within the constraints of a low-cost embedded system.

4.1 Embedded Control Systems Overview

Embedded control systems are specialized computing systems designed to perform dedicated control tasks within a larger mechanical or electrical system. These systems typically operate under real-time constraints and interact directly with sensors and actuators. In incubation applications, embedded controllers are responsible for continuously monitoring environmental conditions and executing control actions with predictable timing and behavior.

4.2 Feedback Control and Hysteresis Logic

Basic feedback control is widely used in environmental control systems due to its simplicity and robustness. In this project, hysteresis-based on/off control was selected to regulate temperature and humidity. Instead of switching actuators exactly at the target setpoint, hysteresis introduces a tolerance band that reduces rapid switching and improves system stability. This approach is particularly suitable for systems using relays and low-cost sensors.

4.3 Non-Blocking Software Architecture

Non-blocking software architecture is a key requirement for real-time embedded systems that manage multiple tasks concurrently. Blocking delays can prevent timely sensor updates and actuator responses. In this project, all time-dependent operations are managed using non-blocking techniques based on system timers, allowing continuous execution of control logic without interrupting other system functions.

4.4 State Machine Design Approach

State machines provide a structured method for modeling system behavior by defining distinct operational states and controlled transitions between them. This approach improves readability, predictability, and maintainability of the control logic. In this project, state machines were used to manage complex operations such as egg turning and ventilation, ensuring that each action follows a predefined and safe execution sequence.

4.5 Safe Boot and Controlled System Initialization

Safe boot behavior is an important design principle in systems that include mechanical motion. Upon startup or reset, the system should avoid initiating any actuator movement without explicit authorization. In this project, the system initializes in a safe idle state, with all actuators disabled. Mechanical actions such as egg turning are only enabled after manual arming by the user, preventing unintended motion after power interruptions.

4.6 Limitations of Mechanical State Retention

Unlike purely digital systems, mechanical systems cannot always preserve their physical state during power loss. Stepper motors, which are commonly used for precise positioning, depend on continuous power to maintain holding torque. In the absence of power, mechanical components may move freely due to external forces such as gravity. This project acknowledges this limitation and avoids attempting to restore mechanical position automatically after power recovery.

5.1 Overall System Architecture

The methodology adopted in this project follows a modular system design approach, where the incubator is divided into sensing, control, and actuation subsystems. The Arduino Mega microcontroller serves as the central control unit, continuously reading sensor data, executing control logic, and driving actuators based on predefined conditions. The system is designed to operate in real time using a non-blocking software structure to ensure continuous monitoring and responsive control.

5.2 Hardware Design

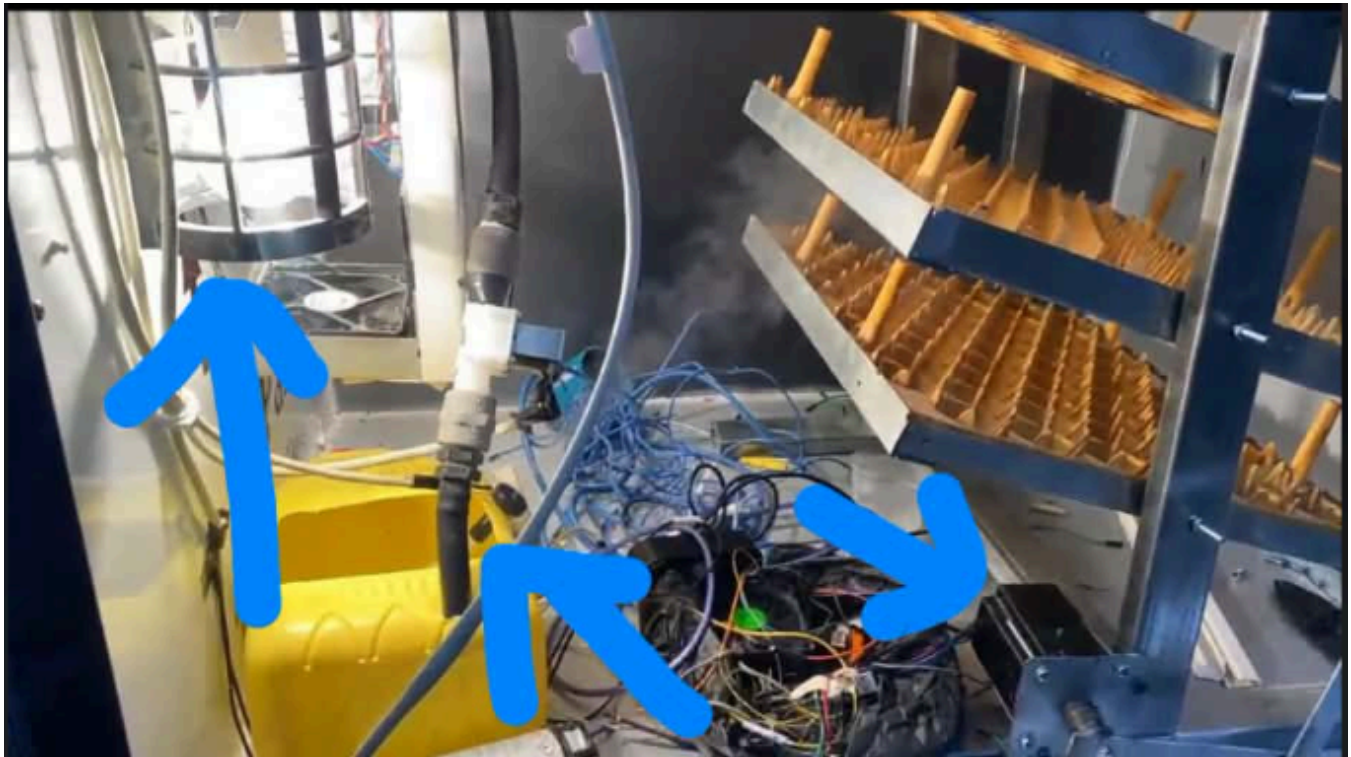
5.2.1 Sensors Selection and Placement

Environmental sensing is a critical component of the incubation process. Temperature and humidity are monitored using a digital sensor placed inside the incubation chamber to accurately reflect internal conditions. Additional sensors are used to detect door status and water level to enhance operational safety and reliability.



5.2.2 Actuators and Mechanical Components

The system employs multiple actuators to control the incubation environment. Heating elements are controlled through relay modules to regulate temperature, while a humidification unit is activated to maintain humidity levels. Ventilation is achieved using a fan and a servo-controlled vent mechanism. Egg turning is performed using a stepper motor driven by a dedicated motor driver, allowing precise and repeatable angular motion.



5.2.3 Power Supply and Electrical Considerations

The system operates using separated power domains to ensure safety and stability. Low-voltage electronics are isolated from high-voltage components controlling heaters and other loads. While no backup power system is implemented, the design prioritizes safe shutdown behavior in the event of power loss by disabling all actuators upon restart.



5.3 Software Design

5.3.1 Main Control Loop Structure

The main control loop is designed using a non-blocking approach, allowing all system tasks to run concurrently. Sensor readings, control decisions, and actuator updates are executed periodically based on system timers rather than blocking delays. This ensures that no single task prevents others from executing in a timely manner.

5.3.2 Environmental Control Logic

Temperature and humidity are regulated using hysteresis-based on/off control logic. Each actuator is activated only when sensor readings exceed predefined thresholds, preventing rapid switching and improving system stability. This method offers a balance between control simplicity and operational reliability.

5.3.3 Egg Turning Control Strategy

Egg turning is implemented using a stepper motor controlled through a state machine-based logic. The turning process is divided into distinct phases, including movement, pause, and return, ensuring smooth and predictable operation. To enhance safety, the turning mechanism remains disabled at startup and is only activated after manual user arming.

5.3.4 Safety-Oriented System Behavior

Rather than relying on active emergency recovery, the system adopts a safety-oriented operational philosophy. In the event of power loss or reset, all actuators are disabled, and the system enters a safe idle state. Mechanical actions are not resumed automatically, and manual user intervention is required to re-enable motion.

5.4 User Interaction and System Control

User interaction with the system is achieved through a simple control interface that allows parameter adjustment and manual arming of the egg turning mechanism. This interface enables controlled operation while preventing accidental activation of mechanical components.



6.1 Microcontroller Platform

Arduino Mega 2560



Definition:

The **Arduino Mega 2560** is the primary microcontroller used in the system. It executes the main control program, continuously reads sensor data, and controls all actuators according to the defined control logic.

Requirement in Project:

The Arduino Mega 2560 was selected due to its large number of digital and analog input/output pins, which is essential for interfacing simultaneously with multiple sensors, relay modules, servo motors, and the stepper motor driver. Its stable performance makes it suitable for handling the real-time control tasks required by the egg incubator system.

ESP32 Module



Definition:

The **ESP32** is a microcontroller module with built-in Wi-Fi and Bluetooth capabilities. It is commonly used for network connectivity and wireless communication in embedded systems.

Requirement in Project:

In this project, the ESP32 module was prepared to enable future network connectivity and remote monitoring capabilities. Communication between the Arduino Mega and the ESP32 is achieved through serial communication. However, advanced networking features such as web-based monitoring and remote control were not fully implemented, as the primary focus of the project was on reliable local control and system stability.

RTC (Real-Time Clock)**Definition:**

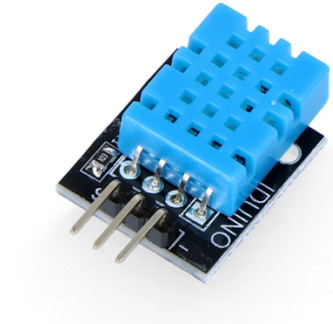
A Real-Time Clock (RTC) is a battery-backed integrated circuit designed to keep accurate time even when the main system power is unavailable.

Requirement in Project:

Although an RTC module was considered during the design phase, it was not integrated into the final system implementation. The system does not require precise absolute timekeeping for its core functionality, as all control operations are based on relative timing intervals. Therefore, system timing was managed using internal software timers, which were sufficient for the operational requirements of the incubator.

6.2 Sensors and Measurement Devices

DHT11 Temperature & Humidity Sensor

**Definition:**

The DHT11 is a digital sensor capable of measuring ambient temperature and relative humidity. It provides calibrated digital output, making it suitable for basic environmental monitoring applications.

Requirement in Project:

In this project, two DHT11 sensors are used. One sensor is placed inside the incubation chamber to monitor the internal temperature and humidity conditions that directly affect embryo development. The second sensor is placed outside the incubator to provide reference environmental data. These measurements are used by the control system to make automated decisions related to heating, humidification, and ventilation.

AngleWater Level Sensor

**Definition:**

A water level sensor is an electronic sensing device that detects the presence or absence of water based on the conductivity between its exposed probes.

Requirement in Project:

In this project, the water level sensor is used to detect low water levels in the humidification system reservoir. When the water level drops below a predefined threshold, the sensor provides a signal to the controller, enabling the system to trigger warnings or prevent improper operation of the humidification mechanism.

Infrared (IR) Sensor**Definition:**

An infrared (IR) sensor is an electronic device that detects the presence or interruption of an infrared signal, allowing simple object detection or proximity sensing.

Requirement in Project:

In this project, the IR sensor is used to detect specific system conditions related to physical access or component positioning. It provides a digital signal to the controller, enabling the system to respond to predefined events and enhance operational awareness.

6.3 Actuators and Output Devices

Servo Motor

Definition:

A servo motor is a compact electromechanical actuator capable of rotating its shaft to a precise angular position, typically within a range of 0° to 180° . It provides accurate position control through an internal feedback mechanism.

Requirement in Project:

In this project, servo motors are used for precise mechanical positioning tasks, such as controlling ventilation openings within the incubator. They enable smooth and repeatable adjustment of airflow by opening or closing vents according to system requirements.

Buzzer**Definition:**

A buzzer is an audio signaling device that produces sound when electrically energized. It is commonly used to provide audible alerts or warnings in embedded systems.

Requirement in Project:

In this project, the buzzer is used as an audible notification device to alert the user of specific system events, such as abnormal conditions, warnings, or user interactions requiring attention.

Ultrasonic Humidifier Maker Fogger**Ultrasonic Humidifier / Humidification Unit****Definition:**

An ultrasonic humidifier is an electronic device that generates fine water mist using high-frequency vibrations. It increases ambient humidity without significantly affecting temperature.

Requirement in Project:

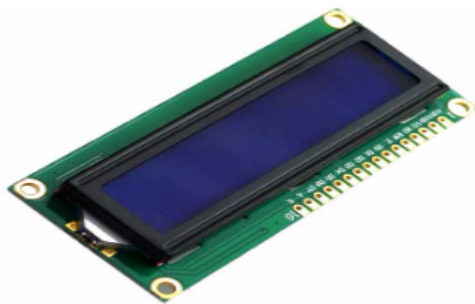
In this project, the humidifier is used to regulate humidity levels inside the incubator. It is activated automatically based on humidity sensor readings and operates in coordination with ventilation to maintain suitable moisture conditions for embryo development.

5V Relay Module (x6)**Definition:**

A relay module is an electrically operated switch that allows a low-voltage control signal to safely switch high-voltage or high-current loads.

Requirement in Project:

Relay modules are used to isolate the low-voltage control circuitry from high-power devices. In this system, relays control heating elements, humidification devices, and ventilation components, ensuring safe and reliable operation of the incubator.

16x2 I2C LCD Display

Definition:

A 16x2 liquid crystal display is a character-based display capable of showing two rows of sixteen characters each. The I2C interface reduces the number of required connections, simplifying hardware integration.

Requirement in Project:

The LCD display serves as the primary local user interface. It provides real-time feedback on temperature, humidity, system status, and user settings without requiring an external computer.

12V DC Water Valve

**Definition:**

A 12V DC water valve is an electromechanical solenoid device that controls the flow of water by opening or closing an internal valve when energized. Unlike a water pump, it does not generate pressure but simply allows or blocks water flow from an external source.

Requirement in Project:

In this project, the water valve is used to control the flow of water supplied to the humidification system. When activated by the Arduino through a relay, the valve allows water to pass to the humidifier or evaporation mechanism, contributing to humidity regulation inside the incubator.

Stepper Motor (Nima 23)

**Definition:**

A stepper motor is a brushless DC motor that divides a full rotation into a fixed number of discrete steps, allowing precise control of angular position and motion without requiring position feedback sensors.

Requirement in Project:

In this project, a NEMA 23 stepper motor is used to drive the egg turning mechanism. The motor provides controlled and repeatable angular movement to tilt or rotate the egg tray according to predefined timing intervals. The stepper motor was selected due to its high torque, positional accuracy, and suitability for slow and controlled mechanical motion.

12V DC Computer Fans (Big Fans)**Definition:**

12V DC computer fans are brushless direct current fans designed to provide continuous airflow with low noise and high reliability. Larger-sized fans offer higher airflow rates, making them suitable for ventilation applications.

Requirement in Project:

In this project, large 12V DC fans are used to provide controlled ventilation inside the incubator. The fans assist in air circulation, heat distribution, and removal of excess humidity. They operate in coordination with the ventilation control logic to ensure a stable incubation environment without causing abrupt temperature or humidity change

4x4 Matrix Membrane Keypad**Definition:**

A 4x4 matrix membrane keypad is an input device consisting of 16 push buttons arranged in a matrix configuration. It allows multiple buttons to be read using a limited number of digital input/output pins by scanning rows and columns.

Requirement in Project:

In this project, the keypad is used as a local user input interface. It allows the user to navigate system menus, adjust operating parameters such as temperature and humidity setpoints, and manually arm or control specific system functions in a controlled manner.

12V Power Supply & 5V Power Supply

**Definition:**

Power supply units provide regulated electrical power required for system operation. A 12V power supply is used to drive higher-power components, while a 5V power supply is used for logic-level electronics and control circuits.

Requirement in Project:

In this project, the 12V power supply is used to power actuators such as fans, motors, and relay coils, while the 5V power supply provides stable power for the microcontroller, sensors, and user interface components. The separation of power levels enhances system safety and reduces electrical interference between control and high-power circuits.

6.4 Development Tools and Software Environment

The software for the system was developed using the Arduino Integrated Development Environment (IDE). This environment provides a simple and effective platform for writing, compiling, and uploading embedded code. Standard libraries were used for sensor interfacing and actuator control, while custom logic was implemented to manage timing, state machines, and non-blocking behavior.

6.5 Hardware Assembly and Prototyping Tools

Hardware assembly and prototyping were carried out using standard electronic tools and components, including breadboards, wiring, connectors, and basic measurement instruments. These tools enabled

flexible testing, iterative development, and rapid debugging during the system integration phase

Chapter 7: Results and Analysis

7.1 System Functional Testing

The system was subjected to a series of functional tests to verify the correct operation of all subsystems. Each component was tested individually before full system integration. Sensor readings were successfully acquired, and actuators responded correctly to control commands. The user interface allowed reliable parameter adjustment and system arming without unintended activation of mechanical components.

7.2 Temperature and Humidity Control Performance

Temperature and humidity control were evaluated under normal operating conditions. The hysteresis-based on/off control logic maintained temperature and humidity within acceptable ranges around the defined setpoints. Minor oscillations were observed, which are expected in systems using relay-based control. However, these oscillations remained within safe limits and did not compromise system stability.

7.3 Egg Turning Performance

The egg turning mechanism was tested over multiple turning cycles. The stepper motor provided smooth and repeatable angular motion, and the state machine-based control ensured consistent execution of movement and pause phases. The manual arming requirement effectively prevented unintended motion during startup or reset, enhancing overall operational safety.

7.4 System Behavior During Power Interruption

Power interruption scenarios were evaluated to observe system behavior under abnormal conditions. Upon power loss, all actuators were immediately deactivated. After power restoration, the system restarted in a safe idle state without initiating any mechanical motion. No automatic resumption of egg turning occurred, requiring explicit user intervention to re-arm the system. This behavior successfully prevented uncontrolled motion after restart.

7.5 Observed Limitations and Practical Issues

During testing, certain limitations were observed. The system does not actively prevent mechanical back-driving when power is lost, as the stepper motor depends on continuous power to maintain holding torque. Additionally, the absence of a real-time clock limits the system's ability to differentiate between short and long power interruptions. These limitations were acknowledged as design trade-offs rather than implementation errors.

7.6 Overall System Evaluation

Overall, the system demonstrated stable and predictable behavior during normal operation and under tested abnormal conditions. The results confirm that the adopted control strategies and safety-oriented design decisions effectively support reliable incubator operation within the defined project scope.

8.1 Interpretation of System Performance

The results obtained from system testing demonstrate that the adopted control strategies achieved the primary objectives of the project. The use of hysteresis-based on/off control provided sufficient stability for temperature and humidity regulation within the incubation environment. Although minor oscillations were observed, they remained within acceptable limits and reflect the expected behavior of relay-driven control systems.

8.2 Evaluation of Egg Turning Strategy

The egg turning mechanism demonstrated reliable and repeatable performance during testing. The use of a stepper motor allowed precise angular control, while the state machine-based implementation ensured smooth sequencing of movement and pause phases. The requirement for manual arming before enabling the turning mechanism proved effective in preventing unintended motion during startup or after system reset.

8.3 Safety-Oriented Design Considerations

A key design decision in this project was prioritizing safety-oriented behavior over automatic recovery. Rather than attempting to restore mechanical states after power interruptions, the system transitions to a safe idle condition and requires explicit user intervention. This approach reduces the risk of uncontrolled motion, particularly in mechanically loaded systems where position retention cannot be guaranteed without power.

8.4 Discussion of Identified Limitations

Despite the successful operation of the system, several limitations were identified. The absence of mechanical locking or braking means that the system cannot prevent back-driving of the egg turning mechanism during power loss. Additionally, the lack of an absolute time reference limits the system's ability to distinguish between different power interruption durations. These limitations highlight areas for future improvement rather than fundamental design flaws.

8.5 Lessons Learned

This project provided valuable practical insights into the interaction between software control logic and mechanical system behavior. It highlighted the importance of designing for predictable and safe operation under abnormal conditions. The experience emphasized that not all system issues can be solved through software alone and that mechanical considerations play a critical role in overall system safety.

9.1 Conclusion

This project presented the design and implementation of a smart egg incubator system based on an embedded control platform. The system successfully integrated environmental sensing, actuator control, and user interaction to provide automated temperature, humidity, ventilation, and egg turning functions. A non-blocking software architecture and state machine-based control logic were adopted to ensure reliable real-time operation.

A key contribution of this project is the emphasis on safety-oriented system behavior. Instead of attempting automatic recovery after power interruptions, the system transitions to a safe idle state and requires explicit user intervention to resume mechanical operation. This design choice reduces the risk of uncontrolled motion and enhances operational safety, particularly in mechanically loaded systems.

Overall, the developed system demonstrated stable and predictable behavior within the defined project scope. The results confirm that a low-cost embedded solution can effectively support automated egg incubation while maintaining reliability and safety.

9.2 Recommendations

Based on the development and testing experience, several recommendations can be made to improve system robustness and functionality. Mechanical enhancements such as counterweights or locking mechanisms could be added to reduce unintended motion during power loss. Improved power conditioning and backup solutions may further enhance electrical stability. Additionally, integrating a real-time clock could improve system awareness of time-dependent events.

9.3 Future Work

Several future enhancements can be explored to extend the capabilities of the proposed system. One significant direction is the integration of a camera-based monitoring system combined with artificial intelligence techniques. By adding a camera module, the system could capture periodic images of the eggs during incubation.

Using computer vision and machine learning algorithms, the system could analyze egg images to differentiate between fertile eggs containing developing embryos and non-viable or damaged eggs. This functionality would allow early detection and removal of spoiled eggs, improving overall incubation efficiency and reducing the risk of contamination.

Additional future work may include advanced control algorithms such as PID control for tighter environmental regulation, data logging for long-term analysis, and network-based monitoring for remote supervision. These enhancements would transform the system from a basic automated incubator into an intelligent incubation platform.

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